Geometric Ergodicity, Uniform Ergodicity, and Tensorisation

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Geometric Ergodicity of Markov Processes

In the study of long-time behaviour of Markov processes, a key property is that of *geometric ergodicity*. Given a Markov semigroup $P = (P_t : t \geq 0)$ with unique invariant measure π, we say that P is geometrically ergodic if there exists a function $M : \mathcal{X} \to [1, \infty)$ and a constant $\lambda > 0$ such that for $x \in \mathcal{X}$ and $t \geq 0$, there holds the bound

$$\mathsf{TV}\left(\delta_x P_t, \pi\right) \leq M\left(x\right) \cdot \exp\left(-\lambda \cdot t\right).$$

This property is somehow critical, in the sense that there are various practically-important features of the Markov chain which hold under the assumption of geometric ergodicity, and which do not hold without it. In coarse terms, it states that convergence to equilibrium takes place at an exponential rate (which is typically the best that one can hope for), but with a 'burn-in' phase to equilibrium which depends on the initialisation x.

Uniform Ergodicity of Markov Processes

A strengthening of this property is that of *uniform ergodicity*, wherein one can upper-bound M uniformly in x, i.e. there exist finite constants $M \geq 1$, $\lambda > 0$ such that for $x \in \mathcal{X}$ and $t \geq 0$, there holds the bound

$$\mathsf{TV}\left(\delta_x P_t, \pi\right) \leq M \cdot \exp\left(-\lambda \cdot t\right).$$

Relative to geometric ergodicity, this property asserts that even for the 'worst-possible' x, this initial burn-in phase cannot be arbitrarily long. As such, this tends to only hold for 'very nice' Markov semigroups, e.g. those defined on compact state spaces, those which make 'global' moves, and so on.

Information Ergodicity of Markov Processes

In other communities, reinforced notions of convergence tend to act by quantifying convergence in different ways, often involving information-theoretic divergences. For example, 'L^2 contraction' refers to the existence of a constant $\lambda > 0$ so that for suitable initial measures μ, there holds the contraction

$$\chi^2\left(\mu P_t, \pi\right) \leq \exp\left(-\lambda \cdot t\right) \cdot \chi^2\left(\mu, \pi\right),$$

where χ^2 denotes the so-called 'chi-squared divergence', and 'entropy contraction' writes as

$$\mathsf{KL}\left(\mu P_t, \pi\right) \leq \exp\left(-\lambda \cdot t\right) \cdot \mathsf{KL}\left(\mu, \pi\right),$$

where KL denotes the 'relative entropy' or 'Kullback-Leibler divergence'. A standard argument shows that entropy contraction implies L^2 contraction, and that this implication cannot be reversed in general.

Tensorisation of Markov Process Convergence

A key aspect of these notions of convergence is that they enjoy a favourable *tensorisation* property. That is, suppose that for $1 \leq i \leq N$, the Markov semigroup $P^{(i)}$ on the space $\mathcal{X}^{(i)}$ has unique invariant measure $\pi^{(i)}$, and satisfies

e.g. L^2 contraction with constant $\lambda^{(i)}$. Define now the tensorised semigroup $P^{[N]} = P^{(1)} \otimes P^{(2)} \otimes \cdots \otimes P^{(N)}$ on the product space $\mathcal{X} = \mathcal{X}^{(1)} \times \mathcal{X}^{(2)} \times \cdots \times \mathcal{X}^{(N)}$ by

$$P_t^{[N]}(\mathbf{x}, \mathrm{d}\mathbf{y}) = \prod_{1 \le i \le N} P_t^{(i)}\left(x^i, \mathrm{d}y^i\right)$$

for each $t \ge 0$, which has unique invariant measure $\pi^{[N]} = \pi^{(1)} \otimes \pi^{(2)} \otimes \cdots \otimes \pi^{(N)}$. It then follows (non-trivially!) that P also satisfies L^2 contraction, and the constant can be taken as $\lambda^{[N]} = \min\left\{\lambda^{(i)} : 1 \le i \le N\right\}$. As a result, one can study high-dimensional chains of product form and obtain dimension-independent rates of convergence. The same phenomenon persists verbatim for entropy contraction.

Quasi-Tensorisation of Markov Process Convergence

Now, returning to geometric ergodicity, one can ask whether the same property holds. The answer is affirmative in some cases, but for slightly indirect reasons: under e.g. reversibility, geometric ergodicity is equivalent to L^2 contraction (even with the same constant), and so one can argue that

$$\mathsf{TV}\left(\delta_x P_t, \pi\right) \le M(x) \cdot \exp(-\lambda \cdot t) \implies \chi^2\left(\mu P_t, \pi\right) \le \exp(-\lambda \cdot t) \cdot \chi^2(\mu, \pi).$$

Tensorisation of L^2 contraction supplies that

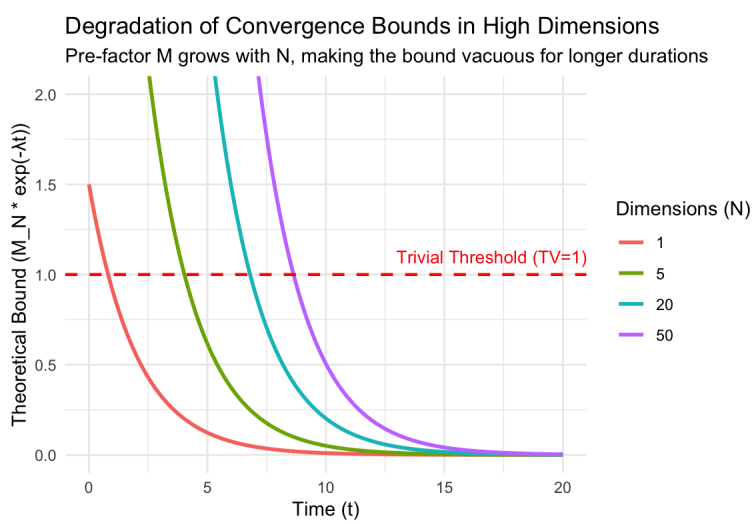
$$\chi^2\left(\mu^{[N]} P_t^{[N]}, \pi^{[N]}\right) \le \exp(-\lambda \cdot t) \cdot \chi^2\left(\mu^{[N]}, \pi^{[N]}\right),$$

and translating backwards yields that

$$\mathsf{TV}\left(\delta_{\mathbf{x}} P_t^{[N]}, \pi^{[N]}\right) \le M^{[N]}(\mathbf{x}) \cdot \exp\left(-\lambda^{[N]} \cdot t\right)$$

for some other compensating pre-factor $M^{[N]} : \mathcal{X}^N \to [1, \infty)$. In the context of uniform ergodicity, the situation is similar; again the rate λ persists, but the constant pre-factor degrades linearly with N.

Figure 1: Large pre-factors delay the convergence upper bounds from being relevant, particularly when bounding bounded divergences like the total variation.



Rehabilitated Tensorisation of Markov Process Convergence by Transport

One reason why these notions of ergodicity behave somewhat poorly in the high-dimensional setting is that they consider multivariate convergence on the same terms as univariate convergence. We will see that by adapting how we quantify convergence, we can obtain much more favourable high-dimensional estimates.

From Discrete Distance to Hamming Distance

Start with uniform ergodicity, i.e. contraction of the total variation distance. It is well-known that the total variation distance can be understood as a transport distance with respect to the discrete metric, i.e. considering $\mathsf{d}(x,y) = \mathbf{1}[x \neq y]$, TV is the 1-Wasserstein distance generated by d.

Taking this metric perspective suggests a new perspective on quantifying convergence in the multivariate case: given $N \in \mathbf{N}$, define for $\mathbf{x}, \mathbf{y} \in \mathcal{X}^N$ the Hamming metric

$$\mathsf{d}^{[N]}(\mathbf{x}, \mathbf{y}) = \sum_{1 \leq i \leq N} \mathbf{1}\left[x^i \neq y^i\right],$$

and instead of considering the total variation distance between high-dimensional probability measures, work instead with the transport distance induced by $\mathsf{d}^{[N]}$, i.e.

$$\mathsf{TV}^{[N]}(\alpha, \beta) = \inf\left\{\mathbf{E}_\gamma\left[\mathsf{d}^{[N]}(\mathbf{X}, \mathbf{Y})\right] : \gamma \in \mathsf{Couplings}(\alpha, \beta)\right\}.$$

That is, instead of assessing convergence on the basis of only exact coalescence of chains, we quantify it on the basis of how many coordinates of the chain have met.

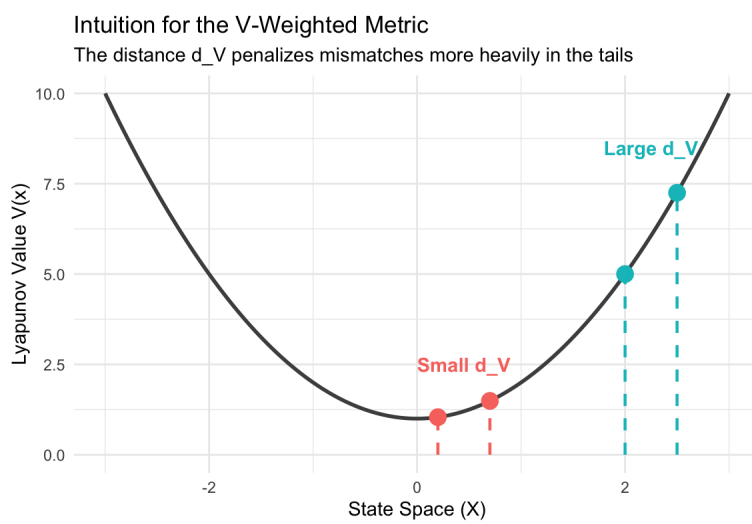
This change of perspective is particularly useful, as metric methods interface especially nicely with high dimensionality. In particular, if $\left\{P^{(i)} : 1 \leq i \leq N\right\}$ are a collection of Markov semigroups which each contract the total variation distance, then the tensorised semigroup $P^{[N]}$ will contract the new distance $\mathsf{TV}^{[N]}$, with the same rate, *and* with the same pre-factor.

Geometric Ergodicity as Metric Contractivity (for some metric)

A similar perspective is available for full geometric ergodicity, though with a somewhat more involved construction. Given a function $V : \mathcal{X} \to [1, \infty)$, define the V-weighted discrete metric by

$$\mathsf{d}_V(x, y) = (V(x) + V(y)) \cdot \mathbf{1}[x \neq y].$$

Figure 2: The weighted discrete metric adjusts its scale to quantify how far the points in question are from 'the center of the state space'.



It is known that in some generality, geometric ergodicity in the form

$$\mathsf{TV}(\delta_x P_t, \pi) \leq M(x) \cdot \exp(-\lambda \cdot t)$$

implies the existence of such a V such that the Markov semigroup P is a contraction on average with respect to the metric d_V, i.e. for some $\mu \in (0, \lambda]$, for arbitrary x, y in $\mathcal{X}$, one can couple draws $X_t \sim \delta_x P_t$, $Y_t \sim \delta_y P_t$ such that

$$\mathbf{E}\left[\mathsf{d}_V(X_t, Y_t)\right] \leq \mathsf{d}_V(x, y) \cdot \exp(-\mu \cdot t).$$

Equally, defining $\mathcal{T}_V$ to be the transport distance associated to d_V, it holds for arbitrary probability measures α, β that

$$\mathcal{T}_V\left(\alpha P_t, \beta P_t\right) \leq \mathcal{T}_V\left(\alpha, \beta\right) \cdot \exp\left(-\mu \cdot t\right).$$

Key elements of this result are due to a contribution of Hairer and Mattingly.

Suppose once more that for $i = 1, 2, \cdots, N$, the Markov semigroup $P^{(i)}$ on the space $\mathcal{X}^{(i)}$ has unique invariant measure $\pi^{(i)}$, and satisfies a geometric ergodicity bound of the form

$$\mathsf{TV}\left(\delta_x P_t^{(i)}, \pi^{(i)}\right) \leq M^{(i)}\left(x\right) \cdot \exp\left(-\lambda^{(i)} \cdot t\right).$$

Following the preceding prescription, each of these guarantees can be translated into a contraction of the form

$$\mathcal{T}_{V^{(i)}}\left(\alpha P_t^{(i)}, \beta P_t^{(i)}\right) \leq \mathcal{T}_{V^{(i)}}\left(\alpha, \beta\right) \cdot \exp\left(-\mu^{(i)} \cdot t\right).$$

Relatively straightforward calculations then establish that the tensorised semigroup satisfies

$$\mathcal{T}_{V^{[N]}}\left(\alpha P_t^{[N]}, \beta P_t^{[N]}\right) \leq \mathcal{T}_{V^{[N]}}\left(\alpha, \beta\right) \cdot \exp\left(-\mu^{[N]} \cdot t\right)$$

with $\mathcal{T}_{V^{[N]}}$ the transportation distance associated to the weighted discrete metric generated by

$$V^{[N]}\left(\mathbf{x}\right) = \sum_{1 \leq i \leq N} V^{(i)}\left(x^i\right).$$

As such, geometric ergodicity can compose nicely in high-dimensional settings, provided that the notions are articulated in an appropriate form.

Conclusion

There are various approaches to quantifying the exponential convergence and contraction properties of Markov processes. For an approach to be well-suited to high-dimensional analysis, it appears that a key structural property is that of tensorisation, i.e. that when studying a system of non-interacting processes, one can easily prove that the composite system equilibrates no more slowly than the slowest constituent process. Popular information-theoretic notions of convergence often satisfy this, as do metric notions of convergence. Some other classical notions of convergence are not so well-suited, but can often be translated into a more amenable form with minimal loss of structural information.